

Elasticity- Solutions

Question-1 (Solution)

- a) The price elasticity of demand (PED) = -0.5556
Since the elasticity of the good is less than 1, the demand for oranges is inelastic.
- b) Total Revenue before price change= \$500
Total Revenue after price change= \$600
With increase in prices, total revenue also increases. This implies that the demand is inelastic.
- c) The income elasticity of demand (YED)=1.24
Since the income elasticity of the good is greater than 0, oranges are a normal good.
And since the elasticity is greater than 1, the demand for oranges is income elastic, indicating that oranges are considered a luxury good.

Question-2 (Solution)

- a) Initial Quantity Demand= 910
- b) Quantity Demand = 935
Price elasticity of Demand (PED)= -0.15
Since $|PED| < 1$, the demand is price inelastic.
- c) Quantity Demand= 925
Cross Elasticity of Demand (XED)=0.074
Since $XED > 0$, ice cream tubs and cones are slightly substitute goods, but the low value means they are weak substitutes.

Question-3 (Solution)

- a) Income Elasticity of Demand (YED) for Good 1 = 1.29 (normal good)
Income Elasticity of Demand (YED) for Good 2= 0.33 (normal good)
Income Elasticity of Demand (YED) for Good 3= -0.60 (inferior good)

Question-4 (Solution)

- a) Price Elasticity of Supply (PES)= 0.65
Since $PES < 1$, the supply is price inelastic.
- b) The quantity supplied responds less than proportionately to the change in price, indicating inelastic supply. This could be because pizzas require time, labour, and fixed oven space making it difficult to increase supply quickly, especially in the short run.